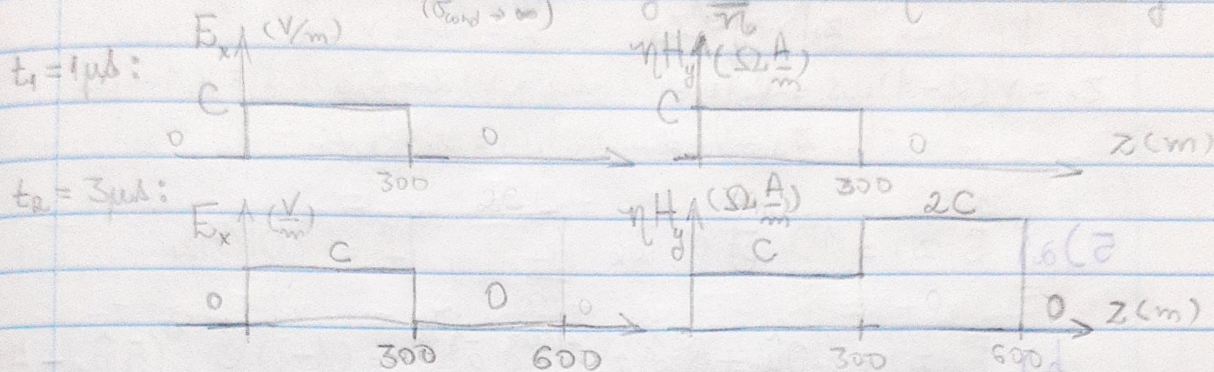


LISTA 3

1) $z_1 = 3 \cdot 10^8 \cdot 10^{-9} = 300 \text{ m}$ $z_2 = 3 \cdot 10^8 \cdot 3 \cdot 10^{-9} = 900 \text{ m}$ $(H \equiv \frac{C}{300})$

$E_x^+ = C \cos(\omega t) = -E_x^-$ $H_y^+ = E_x^+ = C/\eta \cdot u(t) = +H_y^-$

$(\sigma_{\text{cond}} \rightarrow \infty)$ $\eta H_y^+ (\Omega \frac{A}{m})$



2) a) Na superfície: $\vec{J}_s = \hat{u}_n \times (\vec{H}_{y2} - \vec{H}_{y1})$

$-(\hat{u}_z \times \hat{u}_y) 2C \rightarrow \vec{J}_s = +\frac{2C}{\eta} \hat{u}_x$ $\hat{u}_z = \frac{2C}{\eta} \hat{u}_y$

$P_f = \frac{1}{2} |\vec{J}|^2 \eta = \frac{2C^2}{\eta}$

b) $t = 1 \mu s \rightarrow$ ainda não há onda refletida
e $P_f = C^2/\eta$ na região $200 < z < 400 \text{ m}$

c) $z < 300$: $w_E = \frac{1}{2} \vec{D} \cdot \vec{E} = \epsilon \frac{C^2}{2}$; $w_H = \frac{1}{2} \vec{H} \cdot \vec{B} = \frac{\mu C^2}{2} = w_E$
 $\rightarrow$ igualmente distribuída nos campos

$z > 300$: $w_E = 0$ e $w_H = 2\epsilon \frac{C^2}{2}$

A potência em $z = 400$ é nula, armazenada como campo magnético.

3) $f = 1 \text{ MHz}$ $\vec{E} = (4\hat{u}_x + 8e^{j\frac{\pi}{3}}\hat{u}_y + 4\sqrt{3}e^{j\frac{\pi}{2}}\hat{u}_z) \frac{V}{m}$

$\vec{E} = 4 \cdot \cos(\omega t) \hat{u}_x + 8 \cdot \cos(\omega t + \pi/3) \hat{u}_y - 4\sqrt{3} \sin(\omega t) \hat{u}_z$

$= 4 \cos(\omega t) \hat{u}_x + (4 \cos(\omega t) - 4\sqrt{3} \sin(\omega t)) \hat{u}_y - 4\sqrt{3} \sin(\omega t) \hat{u}_z$

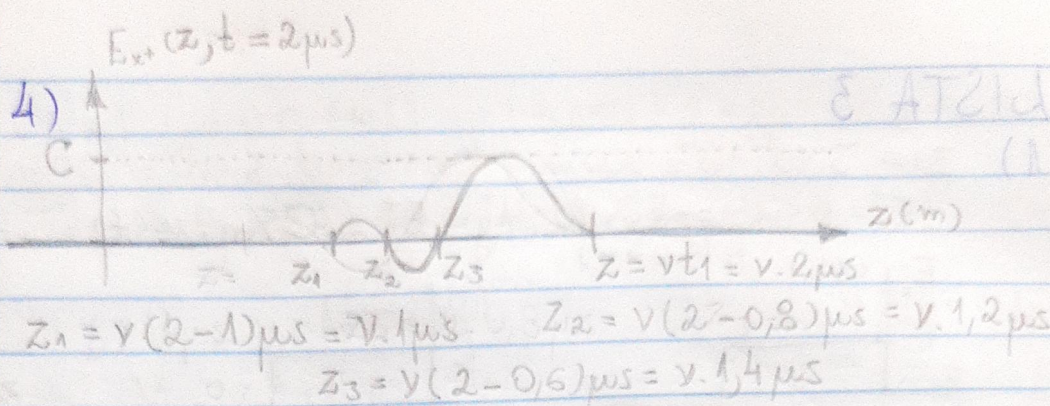
Numa base ortogonal (cosseno e seno) c/ normal ($\hat{n} = c \times s$):

$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 4 & 0 \\ 0 & -4\sqrt{3} & -4\sqrt{3} \end{vmatrix} = 16\sqrt{3} \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$, contida no plano $-x + y - z = 0$, ou $z = y - x$

$|\vec{E}| = \sqrt{16 \cos^2(\omega t) + 64 \cos^2(\omega t + \pi/3) + 48 \sin^2(\omega t)} \rightarrow 4,6549 < |\vec{E}| < 10,3117 \frac{V}{m}$

$= \sqrt{32 \cdot (1 + 2 \sin^2(\omega t) + \sqrt{3} \cos(\omega t) \sin(\omega t))}$

2



5) a) $\vec{E}(z) = \frac{10}{\sqrt{2}} e^{j2z} \hat{u}_x \frac{V_{ef}}{m}$, $f = 100 \text{ MHz}$

b) $\vec{E}(z) = \left(\frac{10\hat{u}_x + 8\hat{u}_y}{\sqrt{2}} \right) e^{j30z} e^{-j\pi/4} = (1-j) e^{j30z} (5\hat{u}_x - 4\hat{u}_y) \frac{V_{ef}}{m}$, $f = 1 \text{ GHz}$

c) $\vec{E}(z) = \frac{(4\hat{u}_x + 3\hat{u}_y)}{\sqrt{2}} e^{j2z} e^{j5x} e^{j\pi/8} \frac{V_{ef}}{m}$, $f = 1 \text{ GHz}$

d) $\vec{E}(z) = \frac{e^{-j\pi}}{\sqrt{2}} (5\hat{u}_x + 10\hat{u}_y) e^{-2z} e^{-j2z} e^{j\pi/8}$

$\vec{E}(z) = \frac{1}{\sqrt{2}} (5\hat{u}_x + 10\hat{u}_y) e^{-2z(1+j)} e^{-j\pi/8} \frac{V_{ef}}{m}$, $f = 100 \text{ MHz}$

e) $\vec{E}(z) = \frac{3}{\sqrt{2}} e^{-2z} \hat{u}_x \cdot e^{j2z} \cdot e^{j\pi/4} = (1,5 + j1,5) e^{-2z(1+j)} \frac{V_{ef}}{m}$, $f = 100 \text{ MHz}$

6) Caso 1: $\gamma k = (1+j) \sqrt{\pi f \mu \sigma} = (1+j) 1560,71 = \alpha + j\beta$
 $\vec{E} = 10 V_{ef}/m$, $\eta = \sqrt{\frac{2\pi f \mu}{\sigma}} / 45^\circ = 35,7727 / 45^\circ \mu\Omega$

a) $\vec{H} = \frac{\hat{u}_z \times \vec{E}}{\eta} = 27954,24 \angle -45^\circ \text{ A}_{ef}/m$

b) $\angle H = -45^\circ = -\pi/4 \text{ rad}$

c) $N = \text{Re}\{\vec{E} \times \vec{H}^*\} = \frac{|\vec{E}|^2}{\text{Re}\{\eta\}} = 19766,634 \text{ W/m}^2$

d) $\lambda_0 = c/f = 30 \text{ Km}$, $\vec{E} \approx 0 \cdot e^{+j2,083} = 0$, $\vec{H} \approx 0 \cdot e^{j1,2975} = 0$, $N = 0$

Caso 2: $\eta k = (1+j) 49354 \text{ 1/m}$; $\eta = (1+j) \sqrt{\mu f \pi} = 1/13/45^\circ$

a) $|\vec{H}| = 883,99 \text{ Aef/m}$

b) $-45^\circ = -\pi/4$

c) $N = 625,046 \text{ W/m}^2$

d) $\lambda_0 = 30 \text{ m}$, $\vec{E} \approx 0 \cdot e^{j0,0256} = 0$

Caso 3: $\eta k = (1+j) 527,94$

$\eta = 105,753/45^\circ \mu\Omega$

a) $|\vec{H}| = 9456 \text{ Aef/m}$

b) $-45^\circ = -\pi/4$

c) $N = 6686,4 \text{ W/m}^2$

d) $\vec{E} \approx 0 \cdot e^{j0,5665} = 0$

Caso 4: $\eta k = (1+j) 16694,84$

$\eta = 3,344/45^\circ \text{ m}\Omega$

a) $|\vec{H}| = 299,025 \text{ Aef/m}$

b) $-45^\circ = -\pi/4$

c) $N = 211,44 \text{ W/m}^2$

d) $\vec{E} \approx 0 \cdot e^{j0,0336} = 0$

7) a) $\vec{E}(z,t) = \sqrt{2} \dot{E}_{xo+} e^{-\alpha z} [\cos(\omega t - \beta z) \hat{u}_x - \sin(\omega t - \beta z) \hat{u}_y]$

$z=0, t=0: \vec{E} = \sqrt{2} \dot{E}_{xo+} (\cos(0) \hat{u}_x + \cos(\pi/2) \hat{u}_y)$

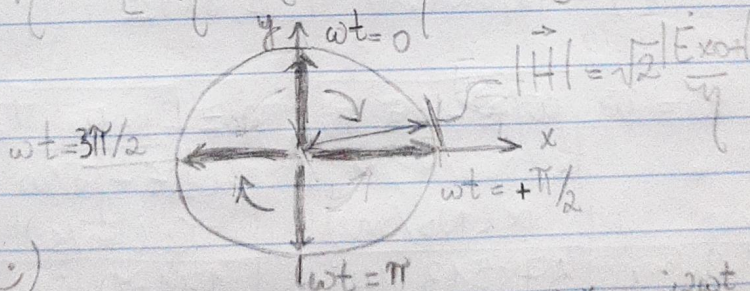
$z=0, \omega t = \pi/2: \vec{E} = \sqrt{2} \dot{E}_{xo+} (0 \hat{u}_x - 1 \hat{u}_y)$

$\omega t = \pi: \vec{E} = -\sqrt{2} \dot{E}_{xo+} (+\hat{u}_x)$

$\omega t = -\pi/2: \vec{E} = +\sqrt{2} \dot{E}_{xo+} (\hat{u}_y)$

$|\vec{E}| = \sqrt{2} |\dot{E}_{xo+}|$

b) $\vec{H} = \frac{1}{\eta} \vec{u}_z \times \vec{E} = \left[\frac{\dot{E}_{xo+}}{\eta} \hat{u}_y + j \frac{\dot{E}_{xo+}}{\eta} (-\hat{u}_x) \right] e^{-j\beta z}$



(circular à esquerda)

c) $N(z=0, t) = \hat{u}_z \{ \text{Re} \{ (\dot{E}_x \dot{H}_y^* - \dot{E}_y \dot{H}_x^*) \} + e^{j2\omega t} (\dot{E}_x \dot{H}_y - \dot{E}_y \dot{H}_x) \}$

$= \text{Re} \left\{ \frac{|\dot{E}_{xo+}|^2}{\eta} - j \frac{2|\dot{E}_{xo+}|^2}{\eta} \right\} + \text{Re} \{ e^{j2\omega t} \cdot 2 \frac{|\dot{E}_{xo+}|^2}{\eta} \} \hat{u}_z$

$= 2 \cdot \frac{|\dot{E}_{xo+}|^2}{\eta} [1 + \cos(2\omega t)] \hat{u}_z$

d) $\vec{E}(z=0, t) = E_0 \cos(\omega t) \hat{u}_x$

$\vec{H}(z=0, t) = \frac{E_0}{\eta} \cos(\omega t) \hat{u}_y$

$N(z=0, t) = \frac{E_0^2}{\eta} [1 + \cos(2\omega t)] \hat{u}_z$

4

8. a) $\phi_y - \phi_x = \pi/2 \Rightarrow$ circular à esquerda

b) vácuo $\Rightarrow \eta = \eta_0 = 377 \Omega$

$$\vec{E}_+(z, t) = \sqrt{2} E_0 \cos(\omega t - \beta z) \hat{u}_x - \sqrt{2} E_0 \sin(\omega t - \beta z) \hat{u}_y$$

$$= 2\sqrt{2} \cos(\omega t - \beta z) \hat{u}_x - 2\sqrt{2} \sin(\omega t - \beta z) \hat{u}_y \text{ mV/m}$$

$$\vec{H}_+(z, t) = 5,3\sqrt{2} \sin(\omega t - \beta z) \hat{u}_x + 5,3\sqrt{2} \cos(\omega t - \beta z) \hat{u}_y \text{ } \mu\text{A/m}$$

c) $\vec{N}_{med} = \Re\{\vec{E} \times \vec{H}^*\} = 2 \frac{E_0^2}{\eta} \hat{u}_z = 2,1,2 \hat{u}_z \text{ } \frac{\text{mW}}{\text{m}^2}$

9. $\vec{E}_+(z, t) = 0,2\sqrt{2} \cos(\omega t - \beta z) \hat{u}_x - 0,2\sqrt{2} \sin(\omega t - \beta z) \hat{u}_y$

$$\eta = \eta_0 / \sqrt{\epsilon_r} = 188,50 \Omega$$

$$\vec{H}_+(z, t) = 1,061\sqrt{2} [\sin(\omega t - \beta z) \hat{u}_x + \cos(\omega t - \beta z) \hat{u}_y] \text{ } \frac{\text{mA}}{\text{m}}$$

10. $k = \beta - j\alpha = \omega \sqrt{\mu(\epsilon' - j\epsilon'')} \approx \omega \sqrt{\mu\epsilon'} (1 - j \frac{\epsilon''}{2\epsilon'})$

$$\eta = \sqrt{\frac{\mu}{\epsilon' - j\epsilon''}} \approx \sqrt{\mu/\epsilon'} \quad v_f = \omega/\beta$$

f (Hz)	α (Np/m)	v_f (10^8 m/s)	η (Ω)
10^6	$4,585 \cdot 10^{-7}$	1,8737	235,46
10^8	$6,562 \cdot 10^{-5}$	1,8774	235,92
10^{10}	$2,827 \cdot 10^{-2}$	1,8811	236,38

11. $k = \omega \sqrt{\mu(\epsilon - j\frac{\sigma}{\omega})}$; $\eta = \sqrt{\frac{\mu}{\epsilon - j\sigma/\omega}}$ e $v_f = \frac{\omega}{\beta}$

	α (Np/m)	v_f (m/s)	η (Ω)
Cobre	478513,14	13130,64	0,0117 $\angle 45^\circ$
Água	77,45	30814449,58	36,2 $\angle 20,8^\circ$

12. a) $\eta = (188,36 \angle 0,23^\circ) \Omega$; $k = (2,096 - j0,0105) \text{ rad/m}$

$$\vec{E}_x(z) = 10 e^{-j2,096z} e^{-0,0105z} \hat{u}_x \text{ mVef/m}$$

$$\vec{H}_y(z) = 53,09 e^{-j(2,096z - 0,005)} e^{-0,0105z} \hat{u}_y \text{ } \mu\text{Aef/m}$$

$$\vec{E}_x(z, t) = \sqrt{2} 10 e^{-0,0105z} \cos(\omega t - 2,096z) \hat{u}_x \text{ mV/m}$$

$$\vec{H}_y(z, t) = \sqrt{2} 53,09 e^{-0,0105z} \cos(\omega t - 2,096z) \hat{u}_y \text{ } \mu\text{A/m}$$

(5)

$$b) \vec{N}_{med}(z) = \frac{|E_{x0}|^2}{\eta} e^{-2\alpha z} \hat{u}_z = 0,5309 e^{-0,021z} \hat{u}_z \frac{\mu W}{m^2}$$

$$c) \Delta \vec{N} = \underbrace{N_0 e^{-2\alpha z_1}}_{N_{med}(z_1)} (e^{-2\alpha \Delta z} - 1) \hat{u}_z \frac{\mu W}{m^2} \approx N_{med}(z_1) 2\alpha \Delta z \frac{\mu W}{m^2}$$

$$P_{diss} = \int_V \omega \epsilon'' |E_x|^2 dV = \omega \epsilon'' \underbrace{A}_{\downarrow} \Delta z |E_{x0}|^2 e^{-2\alpha z_1}$$

$$\text{Como } \frac{2\alpha}{\eta} = \frac{0,021}{188,5} = 1,114 \cdot 10^{-4} \frac{Np}{m \cdot \Omega}$$

$$e \quad \omega \epsilon'' = 2\pi \cdot 50 \cdot 10^6 \cdot 0,04 \cdot 8,854 \cdot 10^{-12} = 1,113 \cdot 10^{-4} \frac{S}{m}$$

são aproximadamente iguais, concluir-se que $\Delta \vec{N} \approx P_{diss}$

$$13. a) \delta = (\pi f \mu \sigma)^{-1/2} = 66,085 \mu m$$

$$b) \eta = (368,96 / 45^\circ) \mu \Omega; k = 15131,914 (1-j)$$

$$\vec{H}(z) = 2 e^{-j15132z} e^{-15132z} \hat{u}_y A_{ef}/m$$

$$\vec{E}(z) = 0,738 e^{-j15132z} e^{-15132z} e^{j\pi/4} \hat{u}_x mV_{ef}/m$$

$$\vec{J}(z) = 42,8 e^{-j(15132z + \pi/4)} e^{-15132z} \hat{u}_x k A_{ef}/m^2$$

$$c) i) \vec{N}(z=0) = \Re \{ |\vec{H}|^2 \cdot \eta \} = 1,0436 \hat{u}_z \frac{mW}{m^2}$$

$$ii) \int_0^\infty |\vec{J}|/2 dz = \frac{\delta}{2} 31,582 \hat{u}_z \frac{W}{m^2}$$

$$14. a) \eta = \eta_0; k = 20,958 \text{ rad/m}$$

$$\vec{E}(z, t) = 5\sqrt{2} \cos(\omega t - 20,958z) \hat{u}_x + 5\sqrt{2} \sin(\omega t - 20,958z) \hat{u}_y \frac{V}{m}$$

$$b) \vec{H}(z, t) = 13,26\sqrt{2} [\cos(\omega t - 20,958z) \hat{u}_y + \sin(\omega t - 20,958z) \hat{u}_x] \frac{mA}{m}$$

$$c) \varphi_y - \varphi_x = -\pi/2; |E_x| = |E_y| \Rightarrow \text{circular, à direita.}$$

$$d) \vec{N}_{med}(z) = \Re \{ \vec{E} \times \vec{H}^* \} = (E_x H_y^* - E_y H_x^*) \hat{u}_z$$

$$= (|E_0|^2/\eta - j |E_0|^2/\eta) \hat{u}_z = 2 \cdot \frac{25}{377} \hat{u}_z = 132,626 \hat{u}_z \frac{mW}{m^2}$$

(6)

$$15. a) \gamma = jk = j2\pi 200 \cdot 10^6 \cdot \sqrt{\mu_0 \epsilon_0} = j4,189 \text{ Np/m}$$

$$\eta = \eta_0 = 377 \Omega$$

$$b) \vec{E}_+(z) = 3 e^{-\gamma z} \hat{u}_x + j 3 e^{-\gamma z} \hat{u}_y \text{ Vef/m}$$

$$\vec{H}_+(z) = -j 8 e^{-\gamma z} \hat{u}_x + 8 e^{-\gamma z} \hat{u}_y \text{ mAef/m}$$

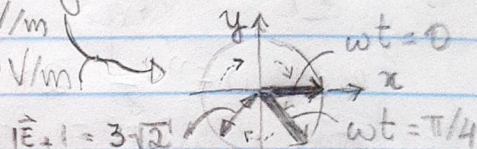
$$c) \vec{E}_+(z, t) = 3\sqrt{2} \cos(2\pi f t - k z) \hat{u}_x - 3\sqrt{2} \sin(2\pi f t - k z) \hat{u}_y \text{ V/m}$$

$$d) \vec{E}_+(z=0, t) = 3\sqrt{2} (\cos(\omega t) \hat{u}_x + \sin(\omega t) \hat{u}_y) \text{ V/m}$$

Que se "distribui" nos vetores ao longo de t ,

$$\text{e vale: } \vec{E}_+(z=0, t=0) = 3\sqrt{2} \hat{u}_x \text{ V/m}$$

$$\text{já } \vec{E}_+(z=0, \omega t = \pi/4) = 3(\hat{u}_x - \hat{u}_y) \text{ V/m}$$



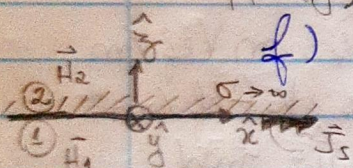
$\varphi_y - \varphi_x = +\pi/2$ e $|E_x| = |E_y| \Rightarrow$ polarização circular à esquerda.

e) Campo elétrico não propaga em condutor ($\rho = -1$)

$$\Rightarrow \vec{E}_-(z) = -3 e^{+\gamma z} [\hat{u}_x + j \hat{u}_y] \text{ Vef/m}$$

$$\vec{H}_-(z) = 8 e^{+\gamma z} [-j \hat{u}_x + \hat{u}_y] \text{ mAef/m}$$

$$f) \vec{J}_s = \hat{u}_z \times (\vec{H}_2 - \vec{H}_1) = (\vec{H}_1(z=0) + \vec{H}_1(z=0)) \times \hat{u}_z = 16 [j \hat{u}_y + \hat{u}_x] \text{ mAef/m}$$



$$16. a) \gamma = j2\pi 10^{10} / c = j209,44 \text{ Np/m}$$

$$\lambda = c/f = 3 \cdot 10^8 / 10^{10} = 3 \text{ cm}$$

b) Circular à esquerda.

$$c) \eta_2 = \sqrt{\frac{\mu_0}{\epsilon_0 - j\sigma/\omega}} = 139,825 / 41^\circ; \rho_0 = 0,583 / 150,52^\circ$$

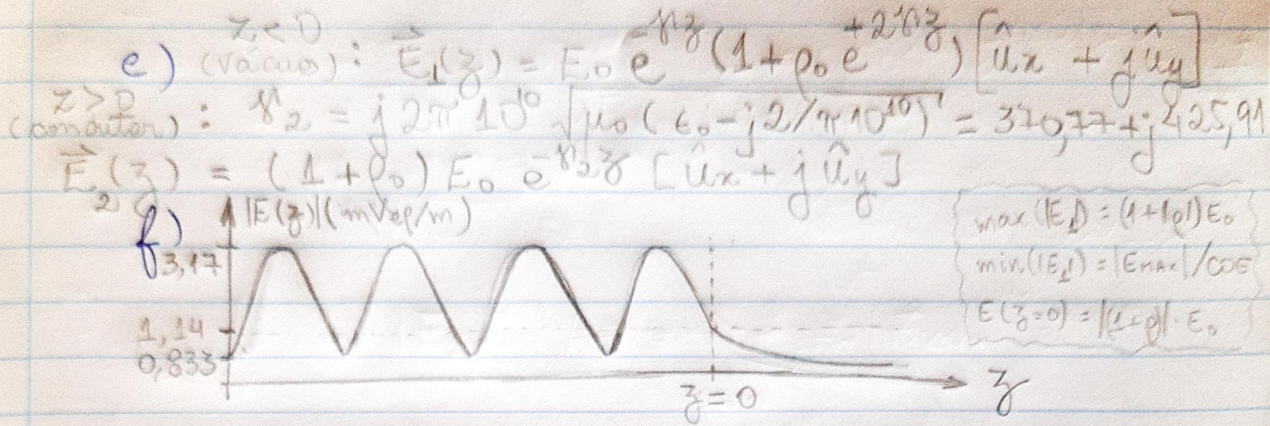
$$\vec{E}_-(z) = \rho_0 E_0 e^{+\gamma z} \hat{u}_x + j \rho_0 E_0 e^{+\gamma z} \hat{u}_y$$

$$= 1,167 \exp\{j(209,44 z + 2,627)\} \cdot [\hat{u}_x + j \hat{u}_y] \text{ mVef/m}$$

$$\vec{E}_-(z, t) = 1,167\sqrt{2} [\cos(20\pi 10^9 t + 209,44 z + 2,627) \hat{u}_x - \sin(20\pi 10^9 t + 209,44 z + 2,627) \hat{u}_y] \text{ V}$$

d) A onda refletida propaga-se no sentido de $-\hat{u}_z$, e nessa referência $\Rightarrow$ polarização circular à direita. Transmitida permanece à esquerda.

(7)



17. a) $\eta_1 = \eta_0 / \sqrt{\epsilon_r} = \eta_0 / 2$; $\rho_0 = -\frac{1/2}{3/2} = -1/3$

$\vec{N}_0 = -N_+ (1 - |\rho_0|^2) \hat{u}_z = 4,716 \hat{u}_z \text{ mW/m}^2$

$\vec{N}_1 = 4,716 \hat{u}_z \text{ mW/m}^2$

b) $k_0 = 2\pi \cdot 3 \cdot 10^9 / c = 20\pi$; $k_1 = 40\pi \text{ rad/m}$

$\vec{E}_0(-0,14) = E_+ e^{-jk_0(-0,14)} (1 + \rho_0 e^{j2k_0 0,14}) = 1,345 / 124,54^\circ \text{ Vef/m}$

$\vec{H}_0(-0,14) = E_+ / \eta_0 e^{jk_0 0,14} (1 - \rho_0 e^{j2k_0 0,14}) = 4,305 / 160,03^\circ \text{ mAef/m}$

$\otimes$

$\vec{E}_1(0,14) = (1 + \rho_0) E_+ e^{jk_1 0,14} = 0,943 / 72^\circ \text{ Vef/m}$

$\vec{H}_1(0,14) = (1 + \rho_0) E_+ / \eta_2 e^{jk_1 0,14} = 5 / 72^\circ \text{ mAef/m}$

c) $\rho(-0,025) = \rho_0 \cdot \exp\{-j2k_0 0,025\} = -\rho_0 = 1/3$

$Z(-0,025) = 754 \Omega$; $Z(0,025) = \eta_1 = 188,5 \Omega$

18. a) $k_1 = 20\pi \sqrt{4 - j0,04} = \pi(40 - j0,2) \frac{\text{rad}}{\text{m}} = \beta_1 - j\alpha_1$

$\eta_1 = \eta_0 / \sqrt{4 - j0,04} = 188,5 / 0,286^\circ \Omega$; $\rho_0 = 1/3 / 179,62^\circ$

$\vec{N}_0 = 4,716 \hat{u}_z \text{ mW/m}^2$; $\vec{N}_1(z) = 4,716 e^{-\pi 0,4z} \hat{u}_z \text{ mW/m}^2$

b) $\vec{E}_0(-0,14) = 1,342 / 124,53^\circ \text{ Vef/m}$

$\vec{H}_0(-0,14) = 0,004312 / 159,974^\circ \text{ Aef/m}$

$\vec{E}_1(0,14) = 0,863 / 72,2^\circ \text{ Vef/m}$

$\vec{H}_1(0,14) = 0,00458 / 71,9^\circ \text{ Aef/m}$

c) $\rho(-0,025) = 1/3 / -0,382^\circ$

$Z(-0,025) = 754,019 / -0,286^\circ \Omega$

$Z(0,025) = 188,4953 / 0,286^\circ \Omega$

8

19. A frequência efetiva que incide no plano é $f' = f(1 + v/c)$ (efeito Doppler).

Igualmente para a onda refletida $f'' = f'(1 + v/c)$ (efetivamente emitida do plano, com velocidade v)

$$f'' = f(1 + v/c)^2 = f(1 + 2v/c + v^2/c^2) \approx f(1 + 2v/c)$$

$$\Delta f = f'' - f = 2f v/c = 2 \cdot 10^{10} \cdot 100 / 1,08 \cdot 10^9 = 1853 \text{ Hz}$$

20. a) $|p| = \frac{\text{CoE} - 1}{\text{CoE} + 1} = 1/2$; $\rho_{\min} = |p| \angle 180^\circ = -1/2$
 $\beta_0 = k_0 = 2\pi \cdot 300 \cdot 10^6 / \lambda$ $\rho(z=0) = \rho_{\min} \cdot e^{+2j\beta_0 \cdot 0,5} = \rho_{\min}$

b) $\eta_{\text{diel}} = \frac{1 + \rho(z=0)}{1 - \rho(z=0)} \eta_0 = \frac{\eta_0}{\text{CoE}} = 125,67 \Omega$

c) Para um dielétrico de pequenas perdas, vale $\epsilon'' \ll \epsilon'$, equivalente a $\sigma \approx 0$, como pode ser observado em η_{diel} - aproximado como real -, e na atenuação $\alpha = 0,001 \ll 1$.

d) $\eta_{\text{diel}} \approx \sqrt{\frac{\mu_0}{\epsilon'}} \Rightarrow \epsilon' = 9\epsilon_0$; $\alpha \approx \omega \sqrt{\frac{\mu_0}{\epsilon'}} \cdot \frac{\epsilon''}{2} \Rightarrow \frac{\epsilon''}{\epsilon_0} = \frac{2}{\epsilon_0} \cdot \frac{\alpha}{\omega \cdot \eta_{\text{diel}}}$

$v_f = \omega / \beta = 10^8 \text{ m/s}$ $\epsilon = (9 - j0,00095) \epsilon_0$

e) $E_{x0-} = \rho(z=0) E_{x0+} = -1 V_{\text{eff}} / \text{m}$

$E_{x0,t} = (1 + \rho(z=0)) E_{x0+} = +1 V_{\text{eff}} / \text{m}$

f) $\vec{N}_{\text{ar}}(z) = \frac{|E_{x+}|^2}{\eta_0} (1 - |\rho|^2) = \frac{|E_{x0,t}|^2}{\eta_{\text{diel}}} = 7,958 \hat{u}_z \text{ mW/m}^2$

$\vec{N}_{\text{diel}}(z) = 7,958 \cdot e^{-0,002z} \hat{u}_z \text{ mW/m}^2$

21. a) $v = \omega / \beta = 400 \cdot 10^6 / 2 = 2 \cdot 10^8 \text{ m/s}$

$\lambda = 2\pi / \beta = 50 \text{ cm}$

Para maiores perdas, o módulo de η_2 diminuiria, e sua fase aproximaria-se de 45° .

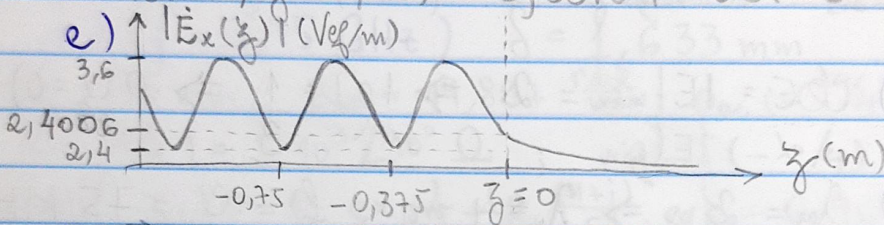
b) $\rho(z=0) = 0,2018 \angle 172,84^\circ \approx -0,2$

$$\vec{E}_{x1}(z=0) = 0,605 / 172,84^\circ \text{ mVef/m}$$

$$\vec{E}_{x2}(z=0) = 2,401 / 1,80^\circ \text{ mVef/m}$$

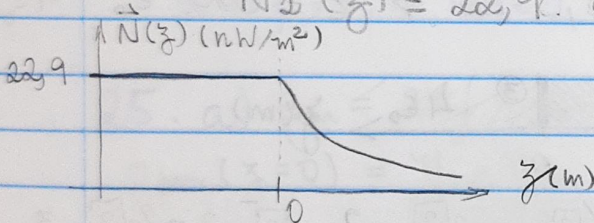
$$\begin{aligned} c) \vec{E}_1(z) &= (3 e^{-j \frac{2\pi}{0,75} z} + 0,605 e^{j(3,017 + \frac{2\pi}{0,75} z)}) \hat{u}_x \text{ mVef/m} \\ \vec{H}_1(z) &= (7,96 e^{-j 8,378 z} - 1,606 e^{j(3,017 + 8,378 z)}) \hat{u}_y \text{ mVef/m} \\ \vec{E}_2(z) &= 2,401 e^{j(4\pi z - 0,0314)} e^{0,63 z} \hat{u}_x \text{ mVef/m} \\ \vec{H}_2(z) &= 14,331 e^{-j(4\pi z + 0,0733)} e^{0,63 z} \hat{u}_y \text{ mVef/m} \end{aligned}$$

$$d) E_x(t, z=1,25) = 0,001545 \cos(\omega t - 3,11017) \hat{u}_x \text{ V/m}$$



$$f) \vec{N}_1(z) = 22,9 \hat{u}_z \text{ nW/m}^2$$

$$\vec{N}_2(z) = 22,9 e^{-1,26 z} \hat{u}_z \text{ mW/m}^2$$



No ar, não há gerador e a potência é constante.

No dielétrico, a potência sofre perdas e aquece o mesmo.

$$22. a) \eta_{\text{diel}} = \eta_0 / \sqrt{\epsilon_r} = 125,67 \Omega; \rho(z=0) = -1/2$$

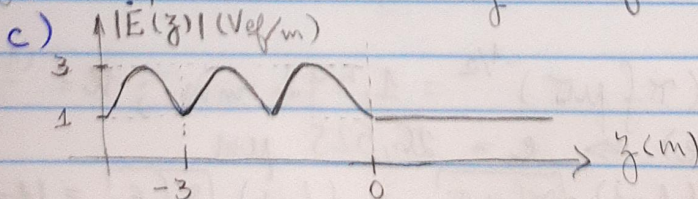
$$b) k_0 = \beta_0 = 2\pi f / c = \frac{2}{3}\pi; k_{\text{diel}} = \beta_0 \sqrt{\epsilon_r} = 2\pi \frac{\text{rad}}{\text{m}}$$

$$\vec{E}_0(z) = (2 e^{-j \frac{2}{3}\pi z} - 1 e^{j \frac{2}{3}\pi z}) \hat{u}_x \text{ Vef/m}$$

$$\vec{H}_0(z) = (5,3 e^{-j \frac{2}{3}\pi z} + 2,65 e^{j \frac{2}{3}\pi z}) \hat{u}_y \text{ mVef/m}$$

$$\vec{E}_1(z) = 1 e^{j 2\pi z} \hat{u}_x \text{ Vef/m}$$

$$\vec{H}_1(z) = 7,958 e^{j 2\pi z} \hat{u}_y \text{ mVef/m}$$



$$d) k_{\text{diel}} \approx 2\pi \cdot 100 \cdot 10^6 \cdot \frac{3}{3 \cdot 10^8} \cdot (1 - j 0,06) = \beta - j\alpha$$

10

$$\alpha = 0,0067\pi \text{ Np/m}; \beta = 2\pi \text{ rad/m}$$

$$\eta_{\text{diel}} \approx \sqrt{\frac{\mu_0}{\epsilon'}} = 125,67 \Omega; v = \frac{2\pi \cdot 10^8}{\beta} = 10^8 \text{ m/s}$$

$$e) \vec{E}_1(z) = 1 \cdot e^{-j2\pi z} \cdot e^{-0,0067\pi z} \hat{u}_x \text{ Vef/m}$$

$$\vec{H}_1(z) = 7,958 \cdot e^{-j2\pi z} \cdot e^{-0,0067\pi z} \hat{u}_y \text{ mAef/m}$$

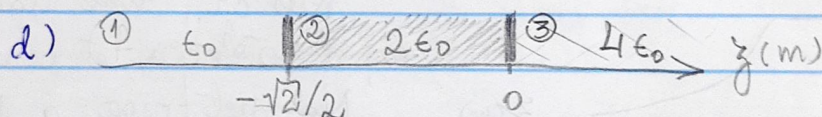
$$f) \vec{N}_{\text{medio}}(z) = 7,958 \cdot e^{-0,013\pi z} \hat{u}_z \text{ mW/m}^2$$

$$g) P_{\text{dis}} = \vec{N}_{\text{medio}}(z=0) \cdot A = 0,00796 \text{ W}$$

$$23. a) \text{COE} = \frac{|E|_{\text{max}}}{|E|_{\text{min}}} = 2 \Rightarrow |\rho| = \frac{1}{3} \Rightarrow \rho(z=0) = -\frac{1}{3}$$

$$b) \frac{\lambda_{\text{ar}}}{2} = 2 \text{ m} \Rightarrow \lambda_{\text{ar}} = 4 \text{ m}; f = \frac{c}{\lambda} = 75 \text{ MHz}$$

$$c) \eta_{\text{diel}} = \frac{\eta_0}{\text{COE}} = 188,5 \Omega \Rightarrow \epsilon = \text{COE}^2 \cdot \epsilon_0 = 4\epsilon_0$$



$$\rho_2(z=0) = \frac{\eta_3 - \eta_2}{\eta_3 + \eta_2} = \frac{1/2 - 1/\sqrt{2}}{1/2 + 1/\sqrt{2}} = \frac{\sqrt{2} - 2}{\sqrt{2} + 2} = \frac{\sqrt{2} - 2}{\sqrt{2} + 2} = 2\sqrt{2} - 3$$

$$\beta_2 = 2\pi f \sqrt{\mu\epsilon} = \frac{2\pi}{4} \cdot \sqrt{2} = \pi \sqrt{2}/2 \text{ rad/m}$$

$$\rho_2(z = -\sqrt{2}/2) = \rho_2(0) \cdot e^{-j2\pi \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2}}{2}} = -\rho_2(0)$$

$$Z(z = -\sqrt{2}/2) = \eta_2 \frac{1 - \rho_2(0)}{1 + \rho_2(0)} = \frac{\eta_0}{\sqrt{2}} \frac{4 - 2\sqrt{2}}{2\sqrt{2} - 2} = \eta_0$$

$$\rho_1(z = -\frac{\sqrt{2}}{2}) = \frac{Z(-\sqrt{2}/2) - \eta_1}{Z(-\sqrt{2}/2) + \eta_1} = 0 \Rightarrow \text{sem reflexão no meio 1}$$

$$24. a) \delta = (\pi f \mu \sigma)^{-1/2} = 1,592 \text{ mm}; k = \frac{(1-j)}{\delta}$$

$$\eta_0 = 1/(\sigma\epsilon) \Rightarrow \epsilon = 26,525 \text{ pF/m}$$

$$k\epsilon = \frac{(1-j)\sqrt{\pi f \mu \sigma}}{\sigma \eta_0} = \frac{(1-j)\sqrt{\pi f \epsilon_0}}{\sqrt{\epsilon_0}} = (1-j)0,0167 \ll 1$$

$$b) \sigma = 100 \gg \omega\epsilon = 2\pi 10^9 \cdot 5\epsilon_0 = 9,278 \text{ S/m}$$

cond. perfeito

$$c) \rho_{\text{ar}}(z=0) = \frac{0 - \eta_0}{0 + \eta_0} = -1; \rho_{\text{ar}}(-\lambda/4) = -1 \cdot e^{-j2 \frac{2\pi}{\lambda} \frac{\lambda}{4}} = +1$$

$$Z(-\lambda/4) = \infty$$

$$\rho_{\text{cond}}(-\lambda/4) = \frac{\infty - \eta_{\text{cond}}}{\infty + \eta_{\text{cond}}} = 1$$

$$\rho_{\text{cond}}(-z_1) = \rho_{\text{cond}}(-\lambda/4) \cdot \exp\left\{-2 \frac{1+j}{\delta} \cdot e\right\} = 0,967 \angle -1,91^\circ$$

$$Z(-z_1) = \frac{1 + \rho_{\text{cond}}(-z_1)}{1 - \rho_{\text{cond}}(-z_1)} \cdot (1+j) \frac{1}{5\delta} = 377 \angle 0,01^\circ \Omega \approx \eta_0$$

$$d) \rho_{\text{ar}}(-z_1) = 0; P_- = |\rho|^2 P_+ = 0 \quad (0\% \text{ reflet.})$$

$$c) (f = 0,95 \text{ GHz}) \quad \delta = 1,633 \text{ mm}$$

$$\lambda_0/4 = 75 \text{ mm}; \beta_{\text{ar}} = 19,897 \frac{\text{rad}}{\text{m}}; \rho_{\text{ar}}(-\lambda_0/4) = 1 \angle 9^\circ$$

$$Z(-\lambda_0/4) = 5 \cdot 10^3 \angle 90^\circ \Omega; \rho_{\text{cond}}(-\lambda_0/4) = +1$$

$$\rho_{\text{cond}}(-z_1) = 1 \cdot \exp\left\{-2 \frac{(1+j)}{\delta} \cdot e\right\} = 0,968 \angle -1,86^\circ$$

$$Z(-z_1) = 377 \angle 0,01^\circ \Omega$$

$$d) (f = 0,95 \text{ GHz}) \quad \rho_{\text{ar}}(-z_1) = 0; P_- = 0\%$$

$$25. a) \beta_{\text{ar}} = 2\pi f/c = 20\pi \frac{\text{rad}}{\text{m}}; \beta_{\text{jam}} = 40\pi \frac{\text{rad}}{\text{m}}$$

$$\rho_{\text{jam}}(z=0) = \frac{\eta_0 - \eta_{\text{jam}}}{\eta_0 + \eta_{\text{jam}}} = \frac{1 - 1/2}{1 + 1/2} = \frac{1}{3}$$

$$\rho_{\text{jam}}(z=-0,1) = 1/3 \cdot \exp\left\{-j2 \cdot 40\pi \cdot 0,1\right\} = 1/3$$

$$Z(-0,1) = 377 \Omega; \rho_{\text{ar}}(-0,1) = 0; E_- = 0\%$$

$$b) \beta_{\text{jam}} = 20\pi \frac{\text{rad}}{\text{m}}; \rho_{\text{jam}}(-0,1) = 1/3$$

$$Z(-0,1) = 377 \Omega; \rho_{\text{ar}}(-0,1) = 0; E_- = 0\%$$

$$c) \beta_{\text{jam}} = 4\pi \text{ rad/m}; \rho_{\text{jam}}(-0,1) = 1/3 \angle -144^\circ$$

$$Z(-0,1) = 110,948 \angle -23,79^\circ \Omega; \rho_{\text{ar}}(-0,1) = 0,58 \angle 14,57^\circ$$

$$E_- = 58\%; P_- = 33,72\%$$

26. $\text{ar} \mid \text{diel. 2} \mid \text{diel. 3} \mid \text{diel. 4} \rightarrow z$

$-d_2-d_3 \quad -d_3 \quad 0$

$$\rho_3(0) = \frac{\eta_4 - \eta_3}{\eta_4 + \eta_3} \rightarrow \rho_3(-d_3) = \rho_3(0) \cdot e^{-j2 \frac{2\pi}{\lambda_3} \frac{\lambda_3}{4}} = -\rho_3(0)$$

$$Z(-d_3) = \eta_3^2 / \eta_4 \rightarrow \rho_2(-d_2-d_3) = -\rho_2(-d_3) = -\frac{Z(-d_3) - \eta_2}{Z(-d_3) + \eta_2}$$

(12)

$$Z(-d_2-d_3) = \frac{\eta_2^2}{Z(-d_3)} = \frac{\eta_2^2}{\eta_3^2} \cdot \eta_4$$

Para casamento perfeito, ρ_1 deve ser zero:

$$\rho_1(-d_2-d_3) = \frac{Z(-d_2-d_3) - \eta_1}{Z(-d_2-d_3) + \eta_1} = 0$$

Ou, $\left(\frac{\eta_2}{\eta_3}\right)^2 \cdot \eta_4 = \eta_1 \rightarrow \frac{\eta_2}{\eta_3} = \left(\frac{\eta_1}{\eta_4}\right)^{1/2}$, Q.E.D.

■ Para frequência 10% abaixo, $\beta z = 0,9 \cdot \frac{2\pi}{\lambda_{10}} = \frac{1,8\pi}{\lambda_{10}}$ para $i \in \{1, 2, 3, 4\}$ e λ_{10} do casamento.

$$\rho_3(0) = -0,172 \rightarrow \rho_3(-d_3) = 0,172 \angle 18^\circ$$

$$Z_3(-d_3) = 185,092 \angle 6,24^\circ \Omega \rightarrow \rho_2(-d_3) = 0,188 \angle 163,76^\circ$$

$$\rho_2(-d_2-d_3) = 0,188 \angle 1,764^\circ \rightarrow Z(-d_2-d_3) = 390,3 \angle 0,69^\circ \Omega$$

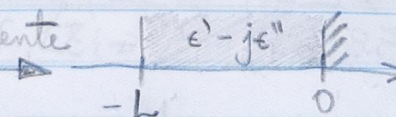
E, finalmente, $\rho_1(-d_2-d_3) = 0,01835 \angle 19,13^\circ$

■ Na figura (b), $\rho_2(0) = \frac{\eta_4 - \eta_2}{\eta_4 + \eta_2} = -1/3$

$$\rho_2(-\lambda_{20}/4) = 1/3 \angle 18^\circ \rightarrow Z(-\lambda_{20}/4) = 360,525 \angle 13,05^\circ \Omega$$

$$\rho_1(-\lambda_{20}/4) = 0,11653 \angle 101,2^\circ$$

Ou seja, o caso em (a) é mais resistente a variações de frequência (reduz as reflexões numa banda maior).

27. Onda incidente 

$$\rho_{\text{diel}}(z=0) = \frac{\eta_{\text{cond}} - \eta_{\text{diel}}}{\eta_{\text{cond}} + \eta_{\text{diel}}} = -1$$

$$\rho_{\text{diel}}(z=-L) = \rho_{\text{diel}}(0) \cdot e^{j2k_{\text{diel}}(-L)} = -e^{-j2k_{\text{diel}} \cdot L}$$

$$Z(-L) = \eta_{\text{diel}} \cdot \frac{1 + \rho_{\text{diel}}(-L)}{1 - \rho_{\text{diel}}(-L)} \quad \begin{cases} k_{\text{diel}} \approx \omega \sqrt{\mu \epsilon'} (1 - j \epsilon''/2\epsilon') \\ \eta_{\text{diel}} \approx \sqrt{\mu/\epsilon'} \end{cases}$$

$$k_{\text{diel}} = 2\pi \cdot 3 \cdot 10^9 \cdot \frac{2}{3 \cdot 10^8} (1 - j \frac{0,01}{4}) = 10\pi (4 - j0,01) \frac{\text{rad}}{\text{m}}$$

$$\eta_{\text{diel}} = \eta_0/2 = 188,5 \Omega$$

$$\rho_{\text{diel}}(-L) = 0,9922 \angle 0^\circ \text{ e } Z(-L) = 48 k\Omega \approx \infty$$

(Poderia, por exemplo, ser notado que $L = \lambda_{\text{diel}}/4$, e, consequentemente, o efeito do condutor aparece na interface dielétrico/vácuo como um aberto $Z: 0 \xrightarrow{\lambda/4} \infty$)

$$28. a) \eta_3 = \sqrt{\frac{\mu}{\epsilon_3}} = \frac{\eta_0}{\sqrt{9}} = 125,67 \Omega; \eta_1 = \eta_0$$

$$k_3 = 2\pi \cdot 10^9 \sqrt{\mu \epsilon_3} = 2\pi \cdot 10^9 \cdot \sqrt{9} = 20\pi \text{ rad/m}$$

$$k_1 = 2\pi \cdot 10^9 / c = 20\pi/3 \text{ rad/m}$$

$$b) \text{ Para } L = \lambda_2/4: \rho_2(-L) = \rho_2(0) e^{-j\pi} = -\rho_2(0)$$

$$\rightarrow Z(-L) = \eta_2 \cdot \frac{1 - \rho_2(0)}{1 + \rho_2(0)} = \eta_2^2 / Z(0) = \eta_2^2 / \eta_3$$

Fazendo máxima incidência de potência (sem

$$\text{reflexão}): Z(-L) = \eta_1 \rightarrow \eta_2 = \sqrt{\eta_1 \cdot \eta_3} = \sqrt{\frac{\eta_0^2}{3}} = \frac{\eta_0}{\sqrt{3}}$$

$$\text{Como } \eta_2 = \eta_0 / \sqrt{\epsilon_2}, \text{ vê-se que } \epsilon_2 = \sqrt{\epsilon_1 \cdot \epsilon_3} = 3$$

$$L = (\underbrace{v_2}_{\lambda_2} / f) / 4 = \frac{c}{\sqrt{\epsilon_2} f} \cdot \frac{1}{4} = 4,33 \text{ cm}$$

$$c) \rho_3(0) = 0 \quad \rho_2(0) = \frac{\eta_3 - \eta_2}{\eta_3 + \eta_2} = -0,268$$

$$\rho_2(-L) = +0,268 \rightarrow Z(-L) = \eta_0 \rightarrow \rho_1(-L) = 0$$

$$d) \bar{N}_1 = \frac{|\vec{E}_1|^2}{\eta_1} (1 - |\rho_1|^2) = \frac{1}{\eta_0} = \bar{N}_3 = \frac{|\vec{E}_3|^2}{\eta_3} (1 - |\rho_3|^2)$$

$$\rightarrow E_3^+ = 1/\sqrt{3} = 0,577 \text{ Vef/m}$$

$$E \left(\frac{\text{Vef}}{\text{m}} \right) \uparrow$$

$$1$$

$$0,577$$

$$-L = -0,0433$$

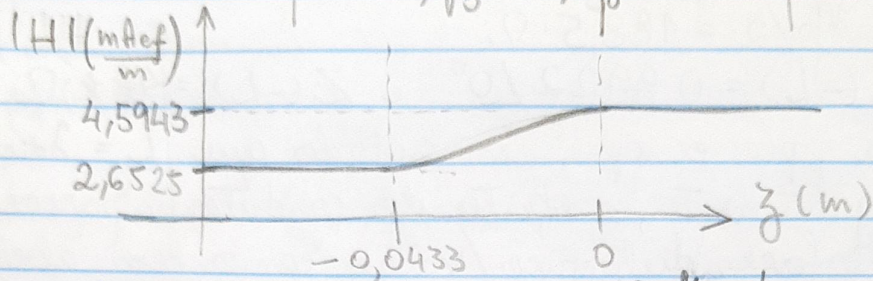
$$0$$

$$z(\text{m})$$

(modo 2, efeito das reflexões)

(14)

e) $\vec{H}_1 = 1/\eta_0 \cdot e^{-jk_1 z} \hat{u}_y \rightarrow |H_1| = 1/\eta_0 = 2,65 \text{ mAef/m}$
 $|H_3| = |\vec{E}_3|/\eta_3 = 1/\sqrt{3} \cdot 3/\eta_0 = \sqrt{3}/\eta_0 = 4,59 \text{ mAef/m}$



f) $\rho_2(-L) = \rho_2(0) \cdot e^{-j2 \cdot k_2 \cdot L}$

Com $k_2 = 2\pi \cdot 2 \cdot 10^9 \cdot \sqrt{3}/c$ e $L = c/(4\sqrt{3} \cdot 10^9)$

ou seja, $\rho_2(-L) = \rho_2(0)$ (pois agora $k = \lambda/2$).

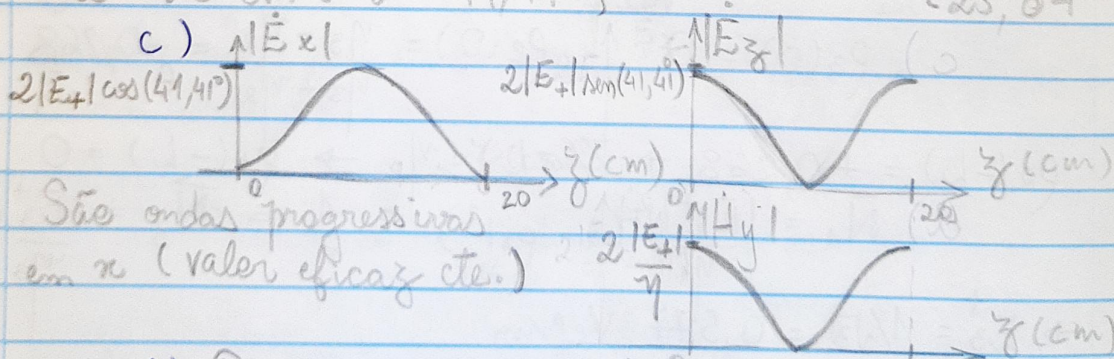
$Z(-L) = \eta_3$ e $\rho_1(-L) = -0,5$

29. a) Transversal magnética (TM)

b) Pelas condições de contorno, impõe-se que $E_x(x, z) = -j2E_+ \cos \theta \sin(kz \cos \theta) e^{-jkx \sin \theta}$

seja zero nas paredes do guia, ou seja $\{n \in \mathbb{Z} \text{ e } k a \cos \theta = n\pi \rightarrow \theta = \cos^{-1}(n\lambda/(2a))\} \{ \lambda = c/f$

• $a = 10 \text{ cm} : \theta \nexists$
 • $a = 15 \text{ cm} : \theta = 0^\circ$
 • $a = 20 \text{ cm} : \theta = 41,41^\circ$
 • $a = 50 \text{ cm} : \theta = \begin{cases} 72,54^\circ \\ 53,13^\circ \\ 25,84^\circ \end{cases}$



São ondas progressivas em z (valor eficaz cte.)

d) Para pequenas distâncias, o guia não é largo o suficiente para permitir a propagação de ondas. Com distâncias maiores, passa a ser possível, com múltiplos ângulos específicos.